

Probability and Bayes' theory

Some definitions

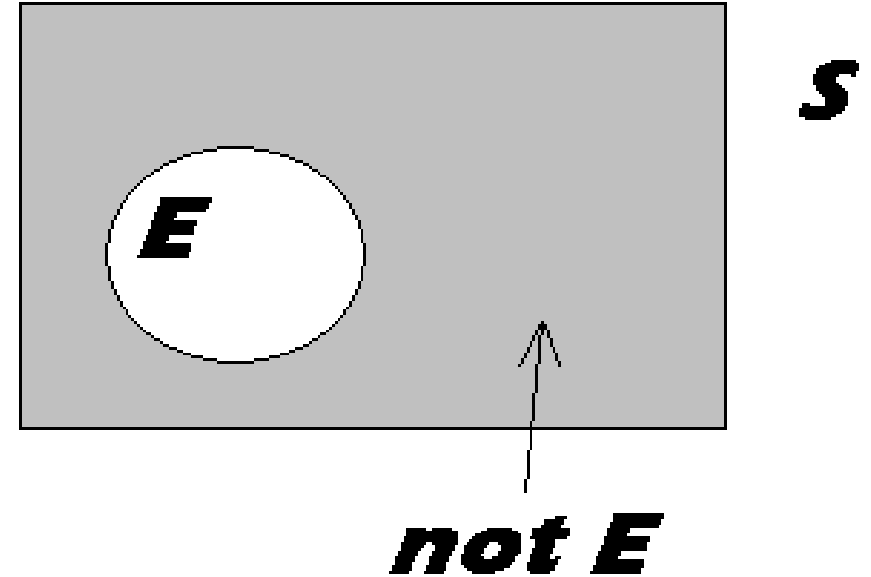
The sample space **S** contains all possible outcomes

An event **E** is a collection of some possible outcomes

E is a subset of **S**

The probability **p(E)** of **E** occurring is a measure of the degree of certainty that **E** occurs

Complements



Rule #1. (Complements)

$$p(\text{E doesn't occur}) = 1 - p(E).$$

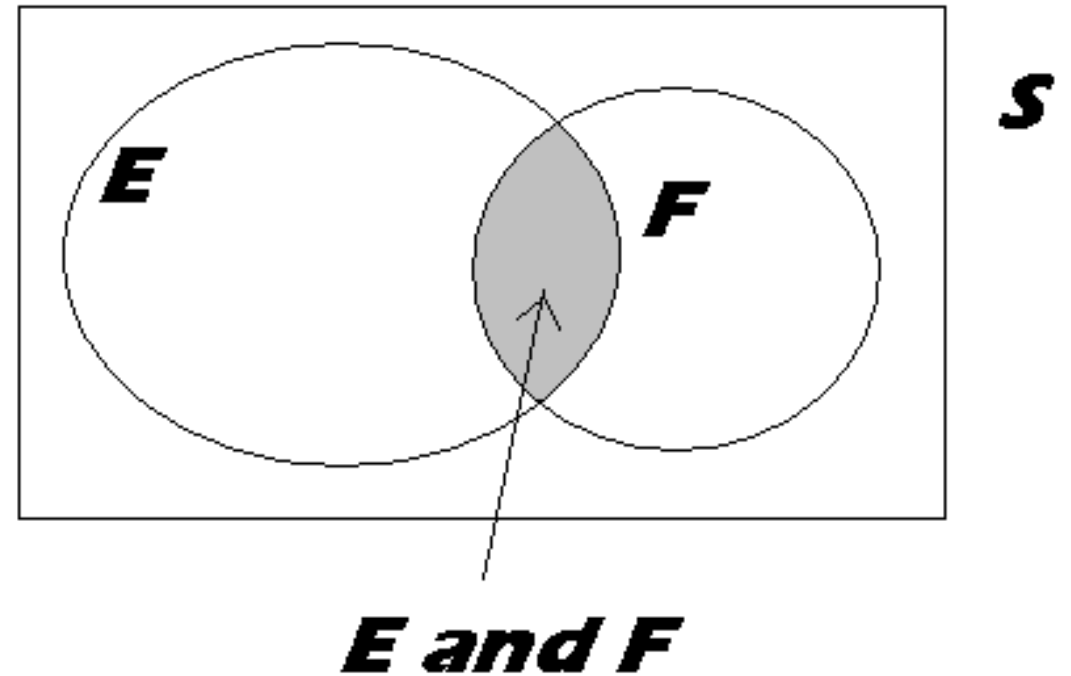
Joint events

Let E and F be events.

The chance that both events occur jointly is the set of all outcomes common to both E and F .

This is the intersection of the two events.

$p(E \cap F)$

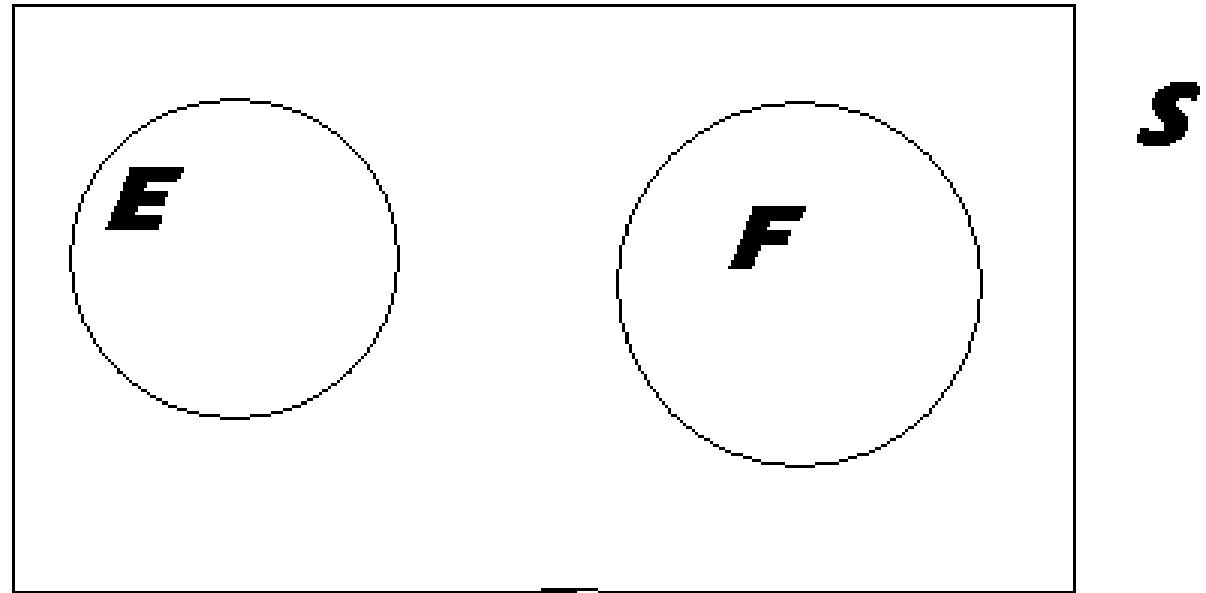


Exclusive events

Rule #2. (Exclusive events)

If E and F are mutually exclusive then:

$$p(E \text{ and } F) = 0.$$



E and F Empty

Combined events

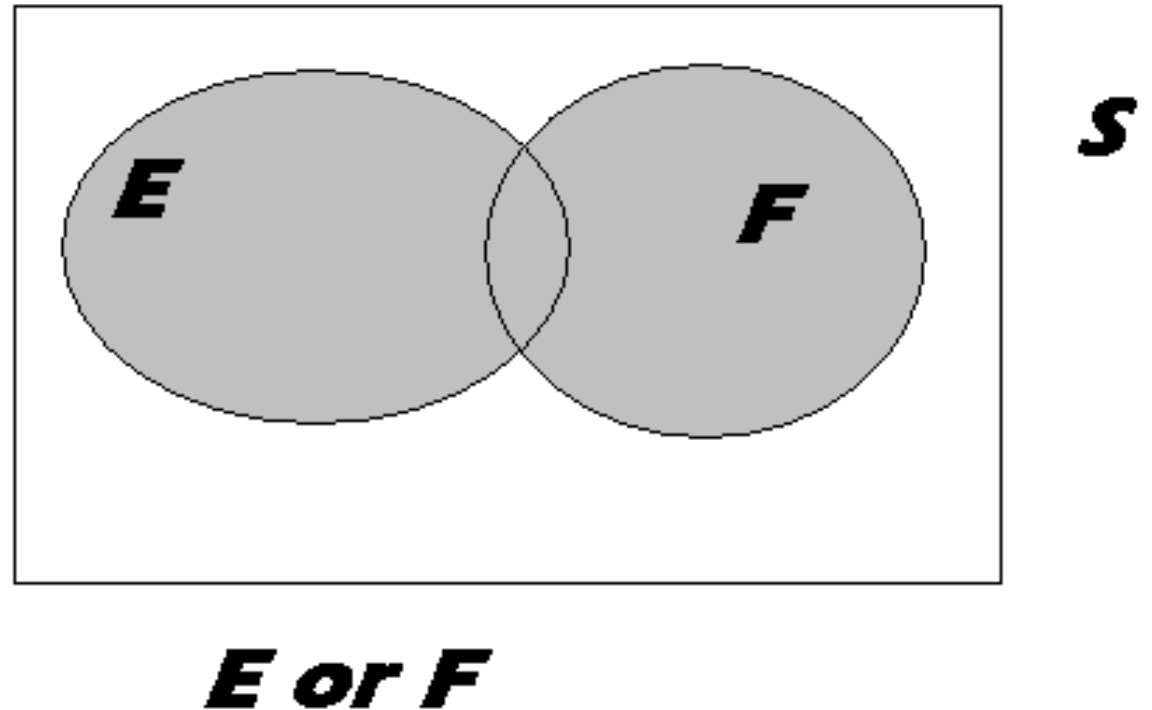
Rule #3. (Joint Probabilities)

$$p(E \text{ or } F) = p(E) + p(F) - p(E \text{ and } F) .$$

In the special case where the events are mutually exclusive, the rule simplifies to

$$p(E \text{ or } F) = p(E) + p(F), \text{ since } p(E \text{ and } F)=0.$$

$$p(E \cup F)$$



Conditional probabilities

Rule #4. (Conditional Probabilities)

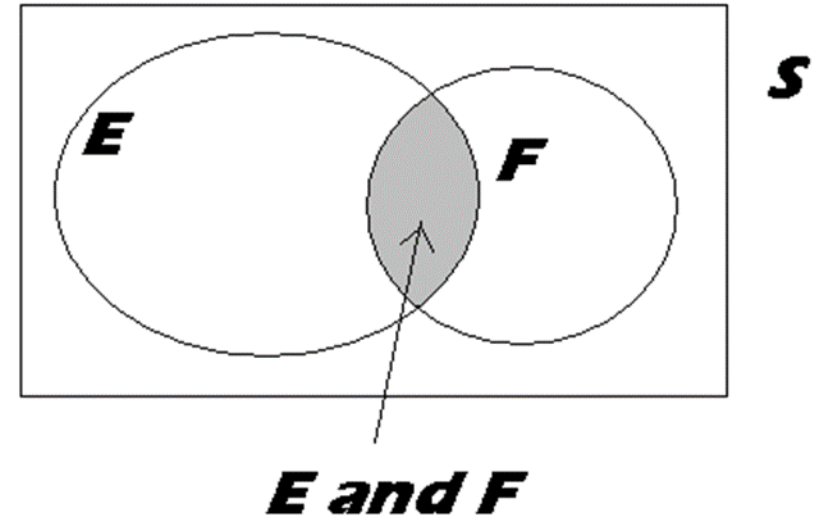
The probability that ***E*** occurs, given the information that ***F*** is known to have occurred, is referred to as a *conditional probability* and will be denoted as by **$p(E|F)$** .

$$p(E|F) = p(E \text{ and } F) / p(F).$$

The formula can be rearranged as:

$$p(E \text{ and } F) = p(F) \times p(E|F)$$

which says that joint probability for E and F to occur is equal to the probability that F occurs times the conditional probability that E occurs given that F is known to occur.



Conditional Joint

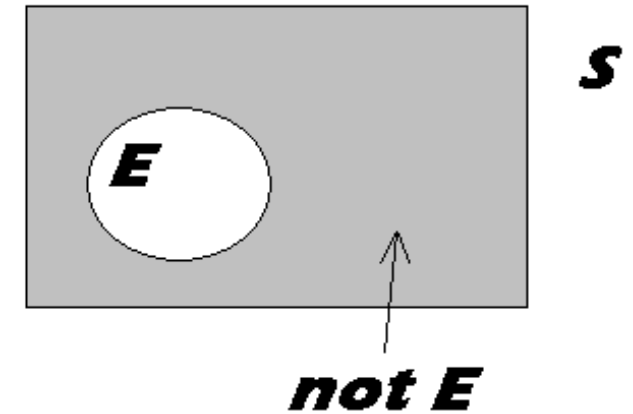
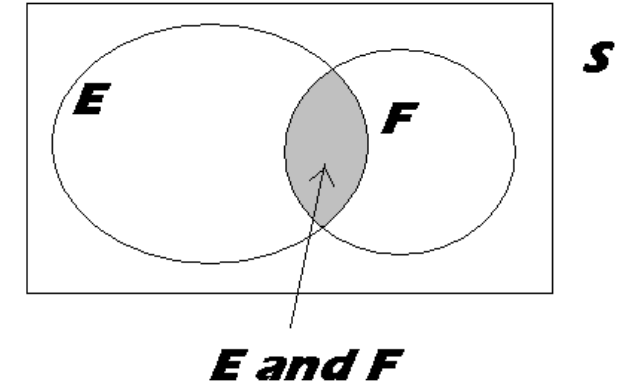
The sum of all probabilities = $S = 1$

$p(E)$ = ratio of area E to S

Conditional probabilities collapse this space down to a new shape (also equal to 1 in area) according to what has already happened

In this case $p(E|F) = p(E \cap F)/p(F)$

F is the new S!



Statistical independence

We say that the events ***E*** and ***F*** are *statistically independent* if we have the following factorisation

$$p(\mathbf{E \text{ and } F}) = p(\mathbf{E}) \times p(\mathbf{F})$$

An immediate consequent of this is that

$$p(\mathbf{E|F}) = p(\mathbf{E}),$$

in other words, the information that F has occurred does not make us change our odds for E to occur.

Statistical independence

Example What are the odds of choosing two hearts from a standard deck of cards if (a) the first card is replaced after selection and the deck reshuffled, (b) no replacement is made?

Let F be the event where the first card is a heart, and E be the event where the second is a heart.

The probability that we want is the joint probability $p(E \text{ and } F)$: that is, both selections to be hearts.

We shall use the formula $p(E \text{ and } F) = p(F) \times p(E|F)$.

To begin with $p(F) = 13/52 = 1/4$.

We now need the conditional probability $p(E|F)$.

Replacement

If we have replacement, then the events E and F are independent.

Therefore $p(E/F) = p(E)$

which again equals $13/52 = 1/4$.

So the answer is $(1/4) \times (1/4) = 1/16$,
or 0.0625 as a decimal.

Independent probabilities

Example:

$$p(\text{sun})=0.4 \qquad p(\text{red car})=0.2$$

$$p(\text{red car \& sun}) = 0.4 \times 0.2 = 0.08$$

$$p(\text{red car} | \text{sun}) = 0.2$$

This is because whether it is sunny or not has no effect on the chance of seeing a red car!

No replacement

Without replacement, the events are **dependent**.

If we know that the first card selected was a heart, then on the second selection we have 12 hearts left from a remaining 51 cards.

Therefore $p(E/F) = 12/51$ and so $p(E \text{ and } F) = (1/4) \times (12/51) = 3/51$, which is approximately 0.0588.

Dependant probabilities

Example:

$$p(\text{sun})=0.4 \qquad p(\text{umbrella})=0.2$$

$$p(\text{umbrella and sun}) = ?$$

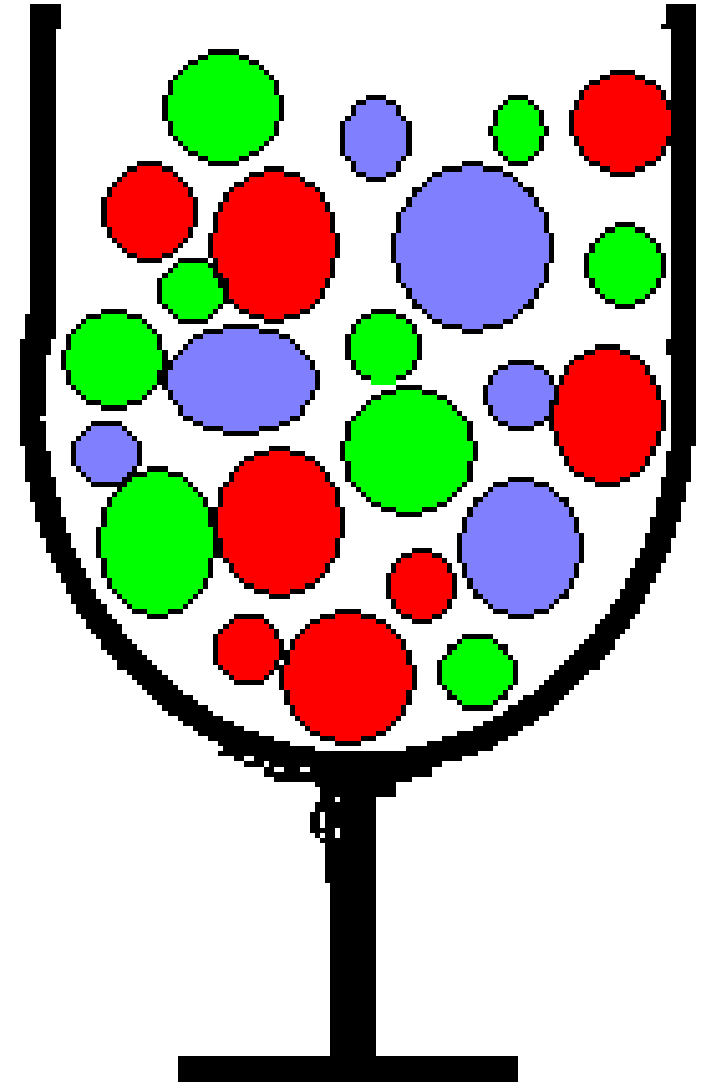
$$p(\text{umbrella} | \text{sun}) = ?$$

Here the weather obviously ***does*** affect the chance of seeing an umbrella!

A worked example

We consider an urn filled with 100 balls of various colours (Red, Green, Blue) and sizes (Large and Small).

In the experiment, we select a ball at random and assume that each of the 100 balls has the same chance ($1/100$) of being the one selected.



Joint probabilities

The chance that the ball selected is both red and large is 5 out of 100. That is $P(R \text{ and } L) = 5/100 = 0.05$. The full set of joint probabilities for this problem are:

$$P(R \text{ and } L) = 0.05$$

$$P(R \text{ and } S) = 0.15$$

$$P(G \text{ and } L) = 0.30$$

$$P(G \text{ and } S) = 0.20$$

$$P(B \text{ and } L) = 0.15$$

$$P(B \text{ and } S) = 0.05$$

	Red (R)	Green (G)	Blue (B)	
Large (L)	5	30	15	50
Small (S)	15	20	15	50
	20	50	30	100

Marginal probabilities

So-called because they use the numbers appearing in the margins!
For instance, the probability that we get a red ball is 20/100. We have:

$$p(R) = p(R \text{ and } L) + p(R \text{ and } S) = 0.05 + 0.15 = 0.20.$$

For the colours, we have

$$p(R) = 0.20$$

$$p(G) = 0.50$$

$$p(B) = 0.30,$$

while for sizes,

$$p(L) = 0.50$$

$$p(S) = 0.50.$$

	Red (R)	Green (G)	Blue (B)	
Large (L)	5	30	15	50
Small (S)	15	20	15	50
	20	50	30	100

Conditional probabilities

Suppose that we have the information that the ball selected is red. We then update our probabilities for the size accordingly.

The conditional probability that the ball is large or small is no longer 50-50, instead, we have:

$$p(L|R) = 5/20 = 0.25$$

$$p(S|R) = 15/20 = 0.75.$$

	Red (R)	Green (G)	Blue (B)	
Large (L)	5	30	15	50
Small (S)	15	20	15	50
	20	50	30	100

Conditional probabilities

Likewise, $p(L|G) = 30/50 = 0.60$, $p(S|G) = 20/50 = 0.40$. The event that we get a large ball is statistically dependent on the event that we get a red or green ball.

	Red (R)	Green (G)	Blue (B)	
Large (L)	5	30	15	50
Small (S)	15	20	15	50
	20	50	30	100

In general, we say that the variables color and size are statistically dependent on each other.

Knowledge of one means that we change our odds for the other.

First worked on by **Thomas Bayes**

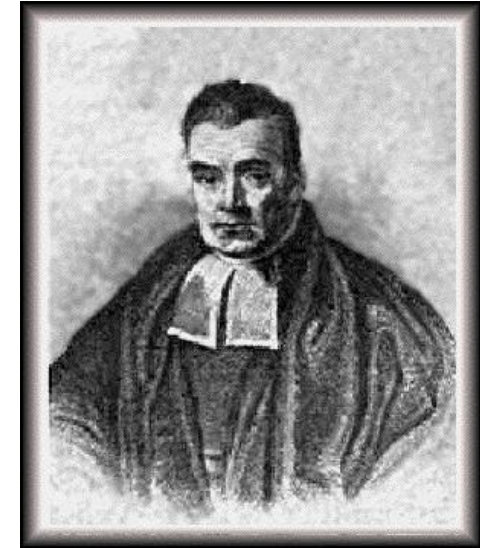
A means of statistical inference

Based on the idea of conditional probability

We examine the evidence for conflicting possibilities and choose to believe the most likely

Allows us to assign probabilities to non-random occurrences

Involves us updating our probabilities as new evidence becomes available



definitions

$p(a)$ is the **marginal** probability of A – the overall chance of A occurring

$p(b)$ is the **marginal** probability of B – the overall chance of B occurring

$p(a|b)$ is the **conditional** probability of A, given B has occurred

$p(b|a)$ is the **conditional** probability of B, given A has occurred

Bayes' theorem

Bayes' theorem relates the two conditional probabilities using the marginal probabilities:

$$P(a | b) = \frac{P(b | a)P(a)}{P(b)}$$

It is not mathematically complicated to use – the important thing is to be **very sure** you have the correct marginal and conditional probabilities in place

An example

Given 100 children in a school, 45 girls and 55 boys; 30 of the boys are dark haired and 36 of the girls are dark haired we can draw up a frequency table:

	boy	girl	
dark hair	30	36	66
light hair	25	9	34
	55	45	100

Marginal probability

In this example it is easy to turn the frequencies into probabilities as we have 100 children over all.

	boy	girl	Marginal P
dark hair	30	36	0.66
light hair	25	9	0.34
Marginal P	0.55	0.45	1

We easily identify the **marginal** probabilities; the chance of randomly selecting someone with dark or light hair or of selecting a boy or a girl.

Conditional probability

We have four conditional probabilities to find:

$$P(\text{boy} | \text{dark}) = 0.3 / 0.66 = \mathbf{0.45}$$

$$P(\text{boy} | \text{light}) = 0.25 / 0.34 = \mathbf{0.74}$$

$$P(\text{girl} | \text{light}) = 0.09 / 0.34 = \mathbf{0.26}$$

$$P(\text{girl} | \text{dark}) = 0.36 / 0.66 = \mathbf{0.55}$$

	P(boy)	P(girl)	
P(dark hair)	0.3	0.36	0.66
P(light hair)	0.25	0.09	0.34
	0.55	0.45	1

If we see a child with light hair there is a 0.26 probability that it is a girl.

However, if we see a girl what is the probability she will have light hair?

$$P(\text{light} | \text{girl})$$

We could calculate again from the table:

$$P(\text{light} | \text{girl}) = 0.09 / 0.45 = 0.2$$

Bayes' theorem allows us another route to this information

$$p(l|g) = \frac{p(g|l)p(l)}{p(g)} = \frac{0.26 \times 0.34}{0.45} = 0.2$$

An intuitive example

The following example is stolen shamelessly from Eliezer Yudkowsky and can be found in full here:

<http://yudkowsky.net/rational/bayes>

1% of women at age forty who participate in routine screening have breast cancer.

80% of women with breast cancer will get a positive mammogram.

9.6% of women without breast cancer will also get a positive mammogram.

A woman of 40 has a positive mammogram in a routine screening.
What is the probability that she has breast cancer?

Decide what your answer would be:

100-80%	Almost certain
80-60%	Fairly likely
60-40%	About half
40-20%	Fairly unlikely
20-0%	Extremely unlikely

When asked; Just **15% of doctors got the answer right!**

In fact we can work it out like this:

In 1000 women aged 40, **10** will have breast cancer. 8 of those have a positive mammogram

Of the 990 women who don't have cancer **95** will also have a positive mammogram (990×0.096)

(Although their chance is lower there are a great many more of them!)

This gives 103 positive mammograms from 1000 ($p=0.103$) and only eight of them actually have cancer!

$$8/103 = 0.078 = 7.8\%$$

We could work this out using the formula also:

$$p(c|m) = \frac{p(m|c)p(c)}{p(m)} = \frac{0.8 \times 0.01}{0.103} = 0.078$$

Interpretation

Conditional probabilities give us information about what has already happened – we can use this information to adjust the probabilities as we know them.

This is an example of the scientific method in that we adjust our view of the world by gathering evidence from experiment.

In this example the knowledge that the mammogram was positive allowed us to adjust our expected risk of cancer.

Alternative formula

$P(m)$ is the probability of a positive mamogram test and is equal to:

$$p(m|c) \times P(c) + p(m|nc) \times p(nc)$$

Where $P(nc) = 1 - P(c) = P(C') =$ Probability of not having cancer. This gives:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A')P(A')}$$

Bayes' theory terminology

$\text{Prob}(a)$ is known as the **prior probability**, since it is the probability measured *prior* to any *new* evidence, data or experience; it is modified by experience.

$$P(a | b) = \frac{P(b | a) P(a)}{P(b)}$$

Experience is expressed through $\text{prob}(b | a)$

$\text{Prob}(a | b)$ is the **posterior probability** – state of belief after *new data* is collected and analysed. i.e. it is the probability that A is true, given that B is true.

$P(b | a)/P(b)$ reflects how evidence of b has caused us to alter $P(a)$

Extension Question: is this true?

$$P(a_1|b) = \frac{P(a_1 \cap b)}{P(a_1 \cap b)P(a_2 \cap b)}$$

Where $P(a_1 \cap a_2)=0$, $P(a_1 \cup a_2) = 1$

More help on probability in Chapter 31, p403